Putting it together: messy data → summary table

Wednesday, June 3, 2026

Today: Wednesday June 3

Before class

- **No new reading** — bring the six verbs from Monday and Tuesday

Plan for today

- No new verbs. Today is about **combining and sequencing** them
- The workflow: question → target table → plan → build → **check**
- Working on a genuinely messy dataset (`flights`)
- Not getting fooled: missing values, tiny groups, wrong order

Helpful links

- Syllabus · AI policy · Schedule · *R4DS* Ch. 3 · *DC* Ch. 7 · *DC* Ch. 10 · dplyr reference

What you already have

Two days in, you have the **whole single-table toolkit**:

Table 1: Six verbs, plus `count()` and the pipe.

Touches…	Verbs
Rows	`filter()`, `arrange()`
Columns	`select()`, `mutate()`
Groups	`group_by()`, `summarise()`

> **Tip**
>
> Today there is **nothing new to memorize.** The skill we're building is *choosing which verb, in what order*, and *checking that the answer is real.*

The workflow

This is Monday's habit (*DC* §7.1), now run end to end:

1. **State the question** in plain English.
2. **Sketch the target table:** what does one row represent? what columns?
3. **Plan the verbs** as English sentences, one per step.
4. **Build the pipeline** one verb at a time, looking at the output each time.
5. **Check it.** Does the count make sense? Are missing values handled? Does the number pass a smell test?

> **Note**
>
> Step 5 is the one people skip, and it's the one Friday's AI lesson was about. A pipeline that *runs* is not the same as a pipeline that's *right*.

A messier dataset

Meet `flights`

You saw `flights` on HW2 and Friday; it's also the running example in R4DS Ch. 3. Every flight out of NYC in 2013:

```
glimpse(flights)
```

```
Rows: 336,776
Columns: 19
$ year           <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
$ month          <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ day            <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ dep_time       <int> 517, 533, 542, 544, 554, 554, 555, 557, 557, 558, 558, ~
$ sched_dep_time <int> 515, 529, 540, 545, 600, 558, 600, 600, 600, 600, 600, ~
$ dep_delay      <dbl> 2, 4, 2, -1, -6, -4, -5, -3, -3, -2, -2, -2, -2, -2, -
1~
```

```
$ arr_time       <int> 830, 850, 923, 1004, 812, 740, 913, 709, 838, 753, 849,~
$ sched_arr_time <int> 819, 830, 850, 1022, 837, 728, 854, 723, 846, 745, 851,~
$ arr_delay      <dbl> 11, 20, 33, -18, -25, 12, 19, -14, -8, 8, -2, -3, 7, -
1~
$ carrier        <chr> "UA", "UA", "AA", "B6", "DL", "UA", "B6", "EV", "B6", "~
$ flight         <int> 1545, 1714, 1141, 725, 461, 1696, 507, 5708, 79, 301, 4~
$ tailnum        <chr> "N14228", "N24211", "N619AA", "N804JB", "N668DN", "N394~
$ origin         <chr> "EWR", "LGA", "JFK", "JFK", "LGA", "EWR", "EWR", "LGA",~
$ dest           <chr> "IAH", "IAH", "MIA", "BQN", "ATL", "ORD", "FLL", "IAD",~
$ air_time       <dbl> 227, 227, 160, 183, 116, 150, 158, 53, 140, 138, 149, 1~
$ distance       <dbl> 1400, 1416, 1089, 1576, 762, 719, 1065, 229, 944, 733, ~
$ hour           <dbl> 5, 5, 5, 5, 6, 5, 6, 6, 6, 6, 6, 6, 6, 6, 6, 5, 6, 6, 6~
$ minute         <dbl> 15, 29, 40, 45, 0, 58, 0, 0, 0, 0, 0, 0, 0, 0, 0, 59, 0~
$ time_hour      <dttm> 2013-01-01 05:00:00, 2013-01-01 05:00:00, 2013-01-
01 0~
```

Where the mess hides

This is what real data looks like after `read_csv("something.csv")`:

- **336,776 rows, 19 columns** — `select()` and `n()` earn their keep here in a way they didn't on `penguins`.
- **Missing values that mean something:** cancelled flights have `NA` for `dep_delay` and `arr_delay`.
- **Codes, not labels:** `carrier` is `"UA"`, not `"United"`. (Turning codes into names needs a *join*, which is later, so today we'll live with the codes.)

```r
flights |>
  summarise(
    n           = n(),
    missing_dep = sum(is.na(dep_delay))
  )
```

```
# A tibble: 1 x 2
       n missing_dep
   <int>       <int>
1 336776        8255
```

> **Warning**
>
> Over 8,000 rows have a missing departure delay. Every average we compute has to decide what to do with them, or it silently returns `NA`.

Worked example 1

Q: which months had the worst delays?

English plan:

- One row per **month**, with its average departure delay and a count.
- *Group by* month, *then summarise* the mean and `n()`, *then arrange* worst-first.

Target table: `month | avg_delay | n` — twelve rows.

```
flights |>
  group_by(month) |>
  summarise(
    avg_delay = mean(dep_delay, na.rm = TRUE),
    n         = n(),
    .groups   = "drop"
  ) |>
  arrange(desc(avg_delay))
```

```
# A tibble: 12 x 3
   month avg_delay     n
   <int>     <dbl> <int>
 1     7     21.7  29425
 2     6     20.8  28243
 3    12     16.6  28135
 4     4     13.9  28330
 5     3     13.2  28834
 6     5     13.0  28796
 7     8     12.6  29327
 8     2     10.8  24951
 9     1     10.0  27004
10     9      6.72 27574
11    10      6.24 28889
12    11      5.44 27268
```

What just happened

We used **three** verbs in sequence, and two safety habits from this week:

- `na.rm = TRUE` so the cancelled flights don't poison the mean *(Monday)*
- `n = n()` so we can see each average rests on ~25,000 flights, not a handful *(Monday)*

> Tip
>
> Summer months (6, 7) and December rise to the top — which matches what you'd expect from thunderstorms and holiday travel. **That agreement is the smell test passing.** If January had come out worst by a mile, you'd go looking for a bug.

Worked example 2

Q: worst JFK summer destinations?

Question: for flights leaving **JFK in summer**, which **destination** had the worst average arrival delay, among destinations with **enough flights to trust**?

English plan:

- *Filter* to JFK, summer months.
- *Group by* destination, *summarise* mean arrival delay + `n()`.
- *Filter again* to drop tiny destinations.
- *Arrange* worst-first.

Build it one verb at a time

```
flights |>
  filter(origin == "JFK", month %in% c(6, 7, 8)) |>
  group_by(dest) |>
  summarise(
    avg_arr_delay = mean(arr_delay, na.rm = TRUE),
    n             = n(),
    .groups       = "drop"
  ) |>
  filter(n >= 100) |>
  arrange(desc(avg_arr_delay))
```

```
# A tibble: 49 x 3
   dest  avg_arr_delay     n
   <chr>         <dbl> <int>
 1 CMH            44.4   194
 2 CVG            39.4   267
 3 IND            30.8   144
 4 PDX            29.6   265
 5 BNA            28.8   184
 6 DEN            28.6   184
 7 CLE            28.2   199
 8 ORF            26.7   128
 9 PBI            25.9   368
10 JAX            22.1   365
# i 39 more rows
```

> **Note**
>
> Notice `filter()` appears **twice**, and it's the *same verb* doing two different jobs: once on the raw rows (which flights), once on the summary table (which destinations are big enough to believe). Same tool, different point in the pipeline.

Why the second filter matters

Without `filter(n >= 100)`, the "worst destination" is whichever one had a single very-late flight. That's exactly Monday's lesson: **a mean from a group of 3 is not a fact.**

```
# Drop the n >= 100 line and the top of the list is full of
# destinations with n = 1 or n = 2. Impressive-looking, meaningless.
```

This is also a verification move: the count column is how you *catch* the problem before it ends up in a report.

Worked example 3

Q: do long hauls leave later?

Now bring in `mutate()` and end with a plot.

English plan: *mutate* a long/short label from `distance`, *group by* it, *summarise* the average delay, *then plot* the summary table.

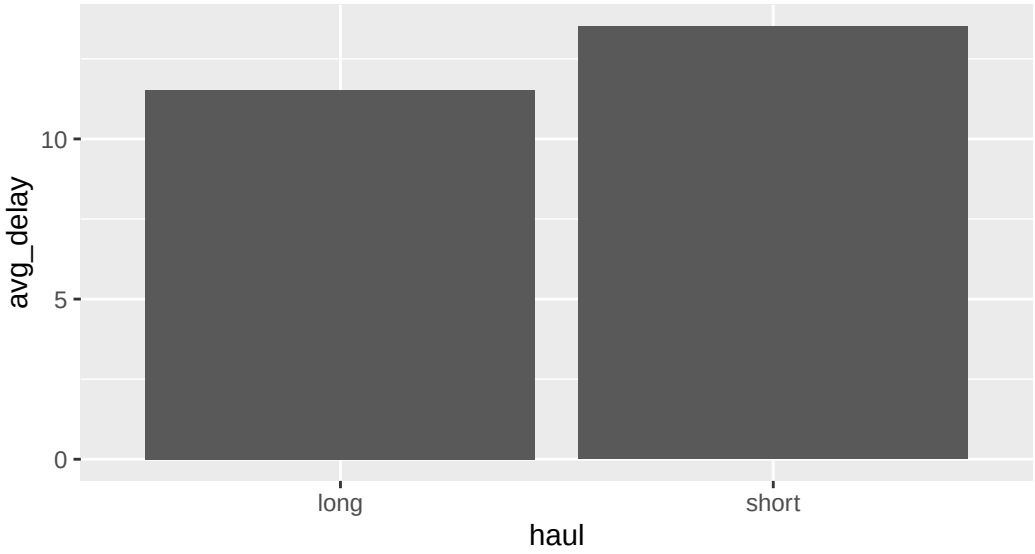
```r
flights |>
  mutate(haul = ifelse(distance > 1000, "long", "short")) |>
  group_by(haul) |>
  summarise(
    avg_delay = mean(dep_delay, na.rm = TRUE),
    n         = n(),
    .groups   = "drop"
  )
```

```
# A tibble: 2 x 3
  haul  avg_delay      n
  <chr>     <dbl>  <int>
1 long       11.5 147105
2 short      13.5 189671
```

From summary table to plot

Because the summary is just a (tiny) data frame, it's already glyph-ready, pipe it into `ggplot()`, switching `|>` to `+`:

```r
flights |>
  mutate(haul = ifelse(distance > 1000, "long", "short")) |>
  group_by(haul) |>
  summarise(avg_delay = mean(dep_delay, na.rm = TRUE), .groups = "drop") |>
  ggplot(aes(x = haul, y = avg_delay)) +
  geom_col()
```



This is the full arc of the week: **raw rows → wrangle to a glyph-ready table → graph it.**

Getting it right

Order matters

The same verbs in a different order answer different questions:

Filter, *then* summarise

"Average delay *of JFK flights.*"

```
flights |>
  filter(origin == "JFK") |>
  summarise(
    avg = mean(dep_delay,
               na.rm = TRUE)
  )
```

Summarise, *then* filter

"Of the per-origin averages, keep JFK's."

```r
flights |>
  group_by(origin) |>
  summarise(
    avg = mean(dep_delay,
               na.rm = TRUE)
  ) |>
  filter(origin == "JFK")
```

Both are valid. The trick is knowing **which question you're actually asking** before you write it.

The three checks

Before you believe any wrangled result, run the checklist that's been building all week:

Table 2: Your verification checklist.

Check	The verb/habit	First seen
Did missing values get handled?	`na.rm`, `!is.na()`	Mon / Tue
Is each number based on enough rows?	`n = n()`	Monday
Does the answer pass a smell test?	compare to what you know	Friday

> **Warning**
>
> A pipeline that runs without error has cleared the *lowest* bar. These three checks are what separate "it ran" from "it's right."

What we did today

- **No new verbs** — just the six from Monday and Tuesday, sequenced
- **The workflow:** question → target table → English plan → build incrementally → check
- **Real mess:** missing delays, code-not-name columns, tiny groups
- **`filter()` in two roles:** subsetting raw rows *and* trimming a summary table
- **Order matters:** filter-then-summarise summarise-then-filter
- **The full arc:** raw rows → glyph-ready summary → plot

Activity

Activity: a flights investigation

Roughly 25 minutes. Individually or with a neighbor.

- Activity Canvas Link

Format: you'll answer a series of questions about `flights` by building pipelines, but each one starts with writing the **English plan first**. You'll compare two pipelines that look alike but answer different questions, catch a silently-dropped `NA`, and finish with one open-ended question where you choose the verbs and **report your sanity checks**.

To turn in: save your `.qmd` and rendered HTML.

Wrap-up & looking ahead

Tomorrow (Thursday): new reading and new content, reshaping data between **wide and narrow** with `pivot_longer()` and `pivot_wider()`.

Upcoming Deadlines

- **Thu 6/4, before class**: *DC* §12.1
- **Thu 6/4, 11:59 p.m.**: Project: Team formation

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

Syllabus · AI policy · Schedule · *R4DS* Ch. 3 · *DC* Ch. 10 · dplyr reference